SOEing PCR protocol to generate KO donors

Description task: Perform 2 PCR reactions to amplify the *galK* upstream and downstream flanking region. In each PCR reaction, a homologous overlapping region (included in tail primer) is introduced to anneal both fragments during the SOEing PCR.

Colony PCR to amplify upstream and downstream homology arms

1. Pick a single BW25113 colony into 50 µL sterile water.
2. Boil at 96°C for 10 min.
3. Spin down briefly in a mini tabletop centrifuge, then use the supernatant as template for the PCR reaction.
4. Mix the following components in a PCR tube for each homology arm (2 tubes total):

Component	**Volume (100 µL)**
10x HF buffer	10 µL
10 mM dNTPs	2 µL
Phusion	1 µL
Primers 10 µM	2 x 5 µL (5 µl FWD, 5µl REV)
Template	10 µL supernatant
Sterile water	67 µL

PCR primers:
Upstream homology arm
FWD: o2514 5' GCAGTCAGCGATATCCATTTTC 3'
REV: o2515 5' GTTTCGTTCACAGGGTAGCCAAATGCGTTG 3'
Tm= 55.6 °C

Downstream homology arm
FWD: o2516 5' GGCTACCCTGTGAACGAAACTCCCGCAC 3',
REV: o2517 5' GACTACCATCCCTGCGTTGTTAC 3'
Tm= 56 °C

5. Place tubes in thermocycler and run program. Note: we used a touchdown PCR (gradual decrease of the annealing temperature in subsequent cycles until it reaches the optimal temperature for primer binding) to minimize non-specific amplification and primer dimer formation, leading to more accurate and efficient PCR results.

Temperature	**Time**	**Number of cycles**
98°C	1 min	Single cycle
98°C	10 s	15 cycles
72°C **(-0.5°C/cycle)**	30 s	
72°C	6 s	
98°C	10 s	15 cycles
Tm°C	30 s	
72°C	6 s	
72°C	5 min	Single cycle

6. Check PCR on 2x gel.

Clean up PCR reaction & determine concentration using Qubit device

Sizes of PCR products will be **97 & 80 bp** for desired fragment. Continue to clean product with 2X ratio magnetic beads. Determine concentration using Qubit device.

Splicing by Overlap Extension PCR

Special two-step PCR to attach two fragments to each other based on homologous overlapping regions. Add **equimolar** concentrations of each fragment to the PCR mix. For SOEing PCR, prepare between 4-6x.

First mix (25µl mix)	
Q5 reaction buffer	5 µl
GC enhancer	5 µl
dNTPs	0.5 µl
Fragment X	x µl
Fragment Y	y µl
Q5 polymerase	0.25 µl
MQ water	14.25 – x – y µl

Note: GC enhancer may not be necessary for different fragments, it is an additive that should be used when dealing with difficult or high GC templates, but can be inhibitory when using high AT content templates.

To calculate x and y (use 10 nM of each fragment):

$$nM_{purified} = \frac{{}^{ng}/_{\mu l}}{650\ {}^{g}/_{mol} * \#bp} * 10^6$$

$$V_{fragment} = \frac{10\ nM * 25\ [\mu l]}{nM_{purified}}$$

Perform 10x PCR cycle with first mix, **use Tm of the overlapping part**. **Note:** no primers added (overlaps serve as primers).

Temperature	**Time**	**Number of cycles**
98°C	3 min	Single cycle
98°C	30 s	10 cycles
59°C	30 s	
72°C	10 s	
72°C	5 min	Single cycle

When the 10x cycles are done, add 25µl of the second PCR mix to each reaction. Use the Tm of the FWD (o2514) & REV (o2517) primer pair and perform 25 cycles.

Second mix (50 µl)	
Q5 reaction buffer	5 µl
GC enhancer	5 µl
dNTPs	0.5 µl
FWD primer (10 µM)	2.5 µl
REV primer (10 µM)	2.5 µl
Q5 polymerase	0.25 µl

sterile water	8.95 µl
First reaction mix	25 µl

Put everything on gel and gel purify (using for example qiagen Qiax protocol). It is very common to get multiple bands, so purification from gel is usually your best option.